MATTHEW PORCELLI

Scottsdale, AZ | 480-212-2274 | MattPorcelli@Protonmail.com
https://mattporcelli.github.io/portfolio/

PROFESSIONAL SUMMARY

Senior Communications/Network Engineer with 20+ years at Cox Communications spanning core MPLS/DWDM network engineering, QoS design and strategy, and network automation. Sets national engineering standards for the Cox Business Network, owns capacity and budget planning, and architects lab automation tooling (Containerlab, Docker, MCP servers) adopted across engineering teams.

CORE COMPETENCIES

Network Design & Capacity Planning • QoS Design & Strategy • MPLS / DWDM / Optical Engineering • Network Automation (Containerlab, Docker, Python) • MCP Server & Tooling Development • Nokia 7x50/7210/7705, Juniper MX, Cisco ASR/CRS • Video Services (Pro Idiom, Contour) • Test Automation (Spirent/Visionworks) • National Market Engineering Governance • Technical Documentation & Standards

PROFESSIONAL EXPERIENCE

Cox Communications (Cox Business) — 2009 – Present

Sr Communications / Network Engineer III — National Engineering Design — 2020 – Present

- **Network Design:** Lead Design Engineer for Nokia Optical projects within the Cox Business Network, nationally; serve as engineering resource for non-BAU network deployments requiring unique capacity or design considerations. Author High-Level Design (HLD) documents for IP/MPLS and DWDM technology evaluations and lab testing initiatives to guide adoption decisions across the Cox Business Network.
- **Budgetary & Capacity Planning:** Lead optical gear lifecycle budgeting, including optical burden calculations across network layers. Own lab budgeting for optical and IP/MPLS lab equipment, driving cost savings by virtualizing lab infrastructure with Containerlab.
- **Automation & Lab Infrastructure:** Architect and maintain Containerlab environments and VM/Docker infrastructure for network testing and R&D, deploying gNMIc, Grafana, and Prometheus for lab telemetry collection and visualization. Maintain lab drawings and design documentation for all new and existing lab devices, providing the accurate device inventory that drives lab budgetary and capacity planning.
- **Custom Application Development:** Leveraged AI-assisted development tools to design and build Harbor (GUI test orchestrator for Nokia robot-framework automation), CLAB Orchestrator (drag-and-drop Containerlab topology builder integrating gNMIc/Grafana/Prometheus/Wireshark), a ZTP Orchestrator proof-of-concept (DHCP Option 67-driven zero-touch provisioning), and Endpoint (Nokia WSNOC API-driven hardware lifecycle and budget tracking).
- **MCP Server Development:** Designed and deployed independent, containerized MCP servers integrating Nokia WSNOC, NFM-P/NSP, Elastic, Snowflake, and CB-NEAT.

Sr Communications / Network Engineer III — National Sustaining Engineering — 2018 – 2020

- **QoS Design & Strategy:** Owned QoS design, strategy, and policy for the Cox Business Network, setting standards adopted network-wide; documented standards in the “QoS Standards Guide” and built QoS calculation spreadsheets for reporting.

- **Tier IV Escalation Authority:** Led resolution of major failed-SLA/circuit issues — Sprint (45+ circuits), T-Mobile (85+ circuits), IPVPN MTU failures, multicast QoS policies failing above 2 Gbps, and 7250 IXR frame discards — producing Technical Bulletins and a QoS Decision Tree adopted across markets.
- **Video Product Engineering:** Drove Pro Idiom (multicast video for Cox Business/Hospitality) and Contour for Hospitality (unicast IPTV replacement), engineering the underlay network service delivering both products across the Cox Business Network: developed and lab-tested CPE-less design standards, co-authored national playbooks with Atlanta Video/Metro Engineering and OSS, and led training across all Cox Business markets.
- **RSAAT Test Automation Platform (Visionworks/Spirent):** Owned the platform end-to-end: built Spirent Centralized and MC-LAG connectors for automated circuit test rebuilding, standardized and cleaned up 100+ probe deployments, designed a new 100G-capable Spirent Orchestrator (100 simultaneous tests, workflow automation, standardized across markets), and developed remote packet-capture MOPs used by markets for customer integration troubleshooting.
- **Engineering Standards & Documentation:** Authored and governed engineering standards documentation; launched the CB Engineering Dashboard for organized access by boundary partners. Authored infrastructure (7750 SR-X Deployment Guide), services (QoS Standards Guide, Pro Idiom & Contour playbooks), process (QoS distribution via NSP, Visionworks/Spirent user provisioning), and testing (Y.1564 SOP, remote packet capture, Spirent connectors) documentation.
- **Cross-Market Collaboration:** Led biweekly Sustaining Collaboration calls (California, Central, Southeast regions) and monthly Service Activation & Assurance Testing calls with market FSA/SA SMEs.

Network Operations Engineer II — Southwest Region **2009 – 2018**

- **Architecture/Design Engineering (Arizona & Las Vegas):** Led capacity planning and budgeting for the Nokia Core MPLS network and access-facing MPLS rings; managed baseline configurations across Nokia 7x50/7210 SAS-M/7210 MXP/7210 SAS-K/7705 SAR, Juniper MX80, Cisco 3750, and Accedian MetroNID/MetroNODE platforms.
- **Compliance & Deployment Standards:** Engineered network changes for Granite/Telcordia compliance; created deployment guides (7210 SAS-K, Y.1731, DS3 circuit emulation) and validated changes in a lab environment prior to production rollout.
- **Escalation & RFP Design:** Served as an escalation path for complex internal/external customer issues; design engineer for the Arizona State University and Raiders Stadium (Las Vegas) RFPs.
- **Cisco DWDM Networks (Arizona & Las Vegas):** Led capacity planning and budgeting for 40/80/96-channel lambda networks; designed DWDM circuits through the Metro Core for Cox Business/Cox Communications Video Services; engineered and deployed a 96-channel Cisco Flex Spectrum network in Phoenix Metro (2017).
- **Standards Compliance:** Identified out-of-compliance network devices and produced engineering remediation packages for Operations teams.
- **IP Backbone & Routing:** Configured and supported IP backbone equipment; multi-protocol routing (BGP, ISIS, OSPF, MPLS) across Cisco 7609/7604/CRS/ASR9000, Cisco 3400/3700, JuniperMX80, and Nokia/Alcatel-Lucent 7x50/7250/7210 platforms; MEF circuit types (EPL, EVPL, EP-LAN, EVP-LAN).
- **Voice & TDM Transport:** Configured and troubleshot H.323/SIP voice services (Cisco AS5400/IAD2400, Edgewater ESBCs, Acme SBCs, BroadWorks Softswitch, Nortel DMS-500) and SONET/DWDM transport (Fujitsu Flashwave/Cisco ONS ADMs, Tellabs Titan/NGX cross-connects, SONET DS1–OC192, DWDM 1G–100G).

CERTIFICATIONS

- Nokia Network Routing Specialist I (NRS1) — 2016
- Cisco CCNP Route & Switch (300-115) — 2015
- Cisco CCNA — 2005 (Recertified 2012)

EDUCATION

- Valencia Community College — Associate of Science, Computer Engineering Technology (Cisco Specialization), 2004
- Valencia Community College — Associate of Arts, 2002
- Cisco Networking Academy — Graduated 2004